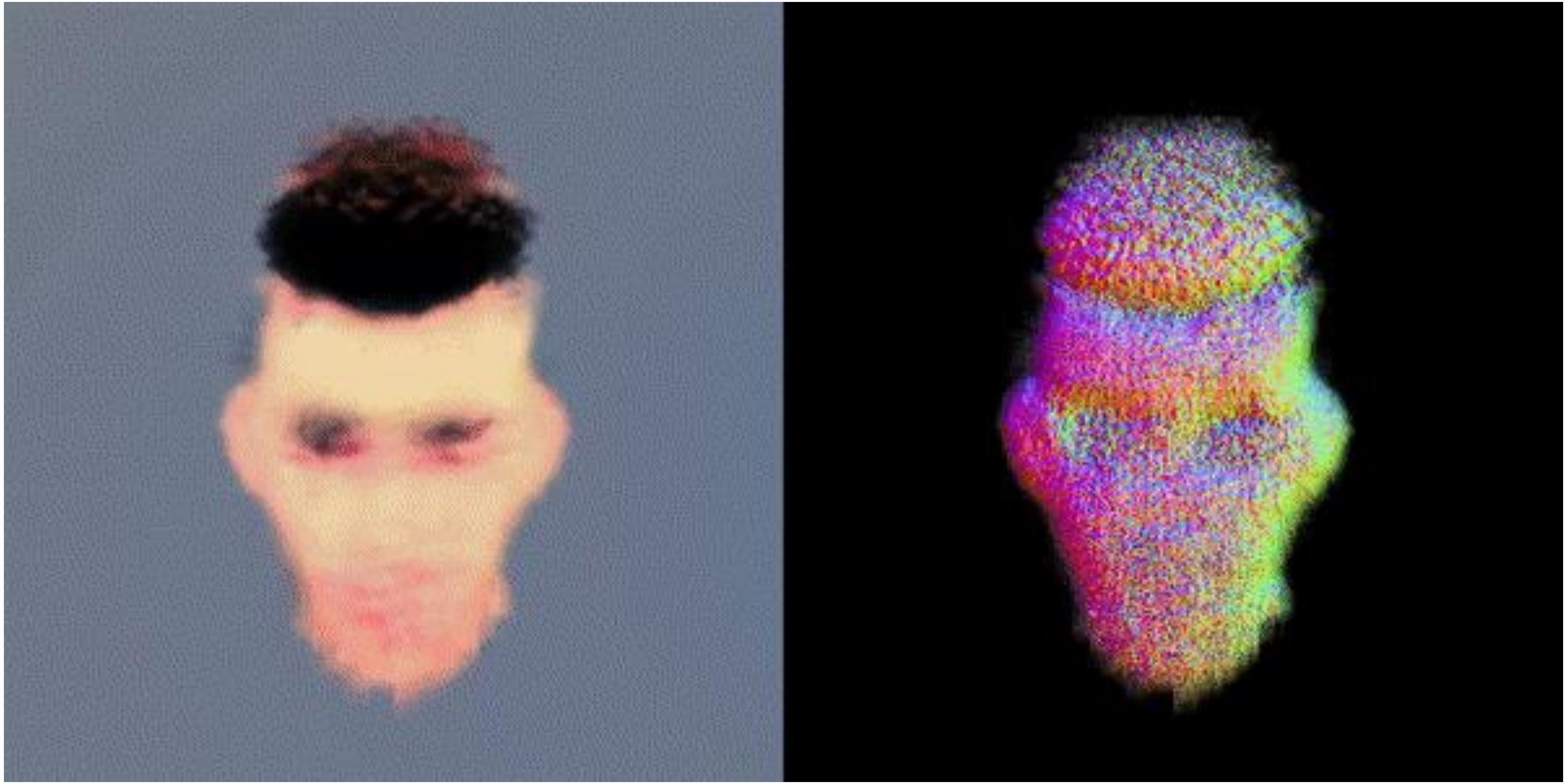
Today's Outline

- Text to 3D Content Generation

DreamFusion, the Text to 3D Framework

Given a caption, DreamFusion generates relightable 3D objects with high-fidelity appearance, depth, and normals. Objects are represented as a Neural Radiance Field and leverage a pretrained text-to-image diffusion prior such as Imagen.

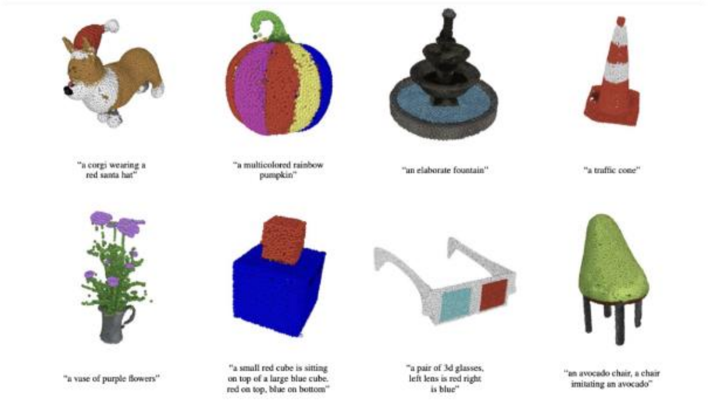
3D Diffusion Models



3D Diffusion Models

Point clouds

- *ShapeGF (Cai et al., 2020)*
- *DPM (Luo and Hu, 2021)*
- *LION (Nichol et al., 2022)*

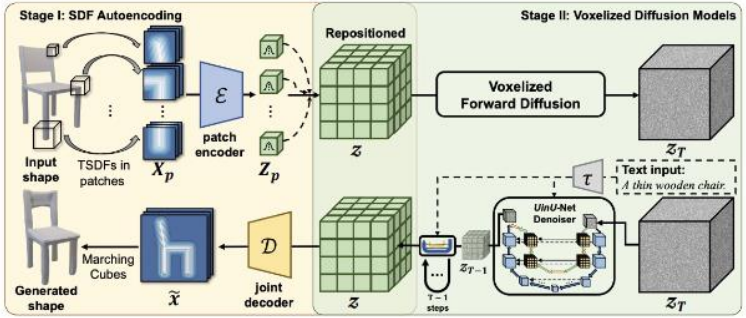


LION, Nichol et al. 2022.

3D Diffusion Models

Voxel representation

- *PVD (Zhou et al., 2022)*
- *Diffusion-SDF (Li et al, 2022)*

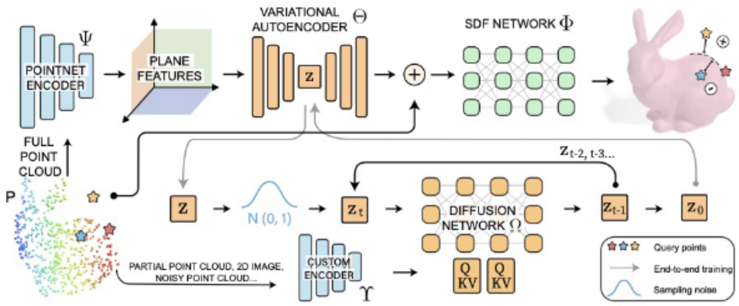


Diffusion-SDF, Li et al., 2022.

3D Diffusion Models

Latent representation

- *SDFusion* *(Cheng et al., 2022)*
- *DiffusionSDF (w/o hyphen, Chou et al., 2023)*



DiffusionSDF, *Chou et al., 2023*

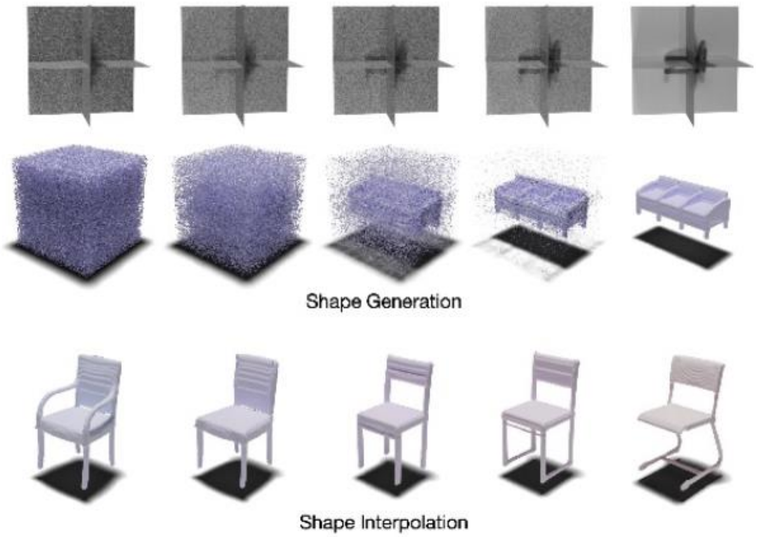
3D Diffusion Models

Triplane representation

- *NFD (Shue et al., 2022)*

Diffusion in the spectral domain:

- *NeuralWavelet (Hui et al., 2022)*



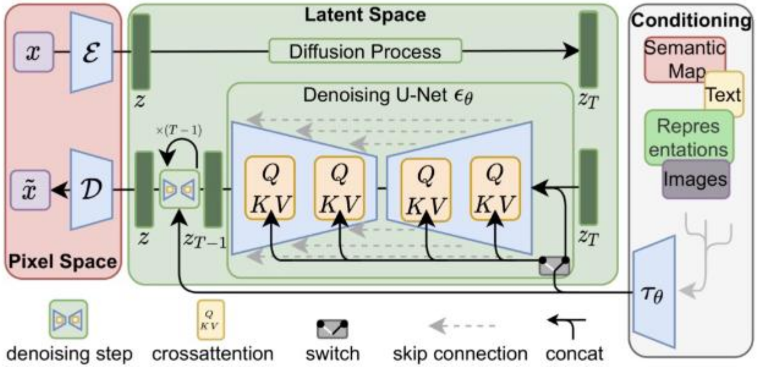
NFD, Shue et al., 2022.

Diffusion w/ Different Representations

- *Implicit representation*
- *Explicit representation*

Diffusion w/ Different Representations

- *Implicit representation (I.e., latent features)*
  - *E.g., Latent diffusion (Rombach et al., 2022)*
  - *(+) Best quality of the generated data.*
  - *(−) Requires retraining for each conditional generation setup.*

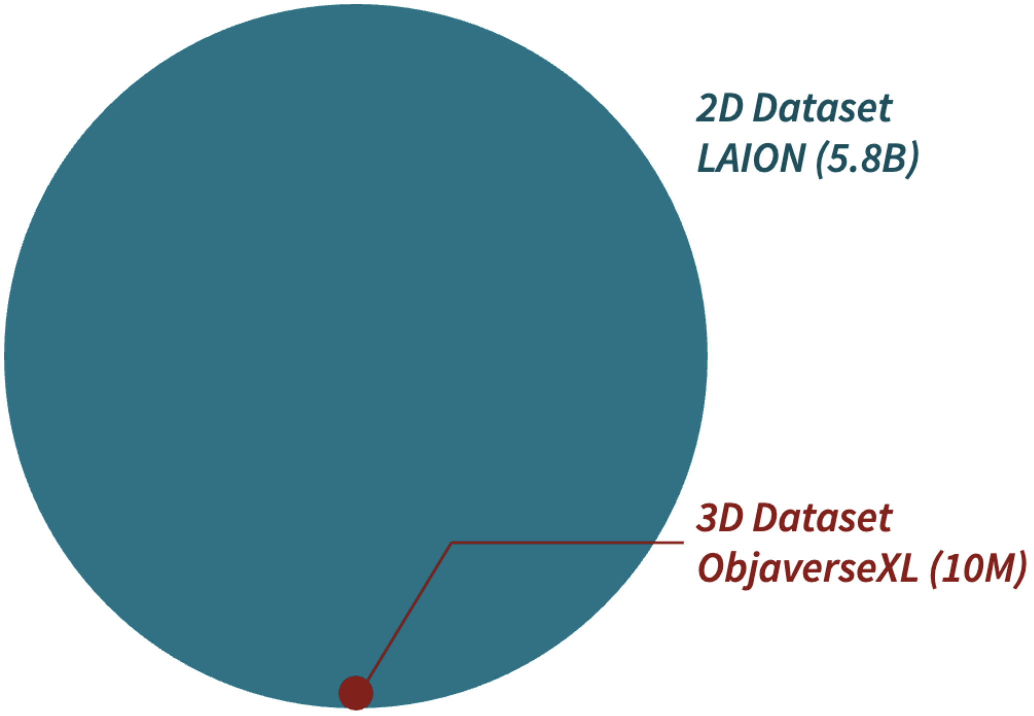


Latent Diffusion, Rombach et al., 2022.

Diffusion w/ Different Representations

- *Explicit representation (E.g., pixels in images)*
  - *E.g.: The original DDPM model (Ho et al., 2020) for images.*
  - *(–) Suboptimal performance due to the high dimensionality.*
  - *(–) Cannot change the resolution of the data.*
  - *(+) Can be directly leveraged in conditional generation setups in a zero-shot manner.*

Data Deficiency



Diversity of Imaginable 3D Shapes

"frog wearing a sweater"

"eggshell broken in two with an adorable chick standing next to it"

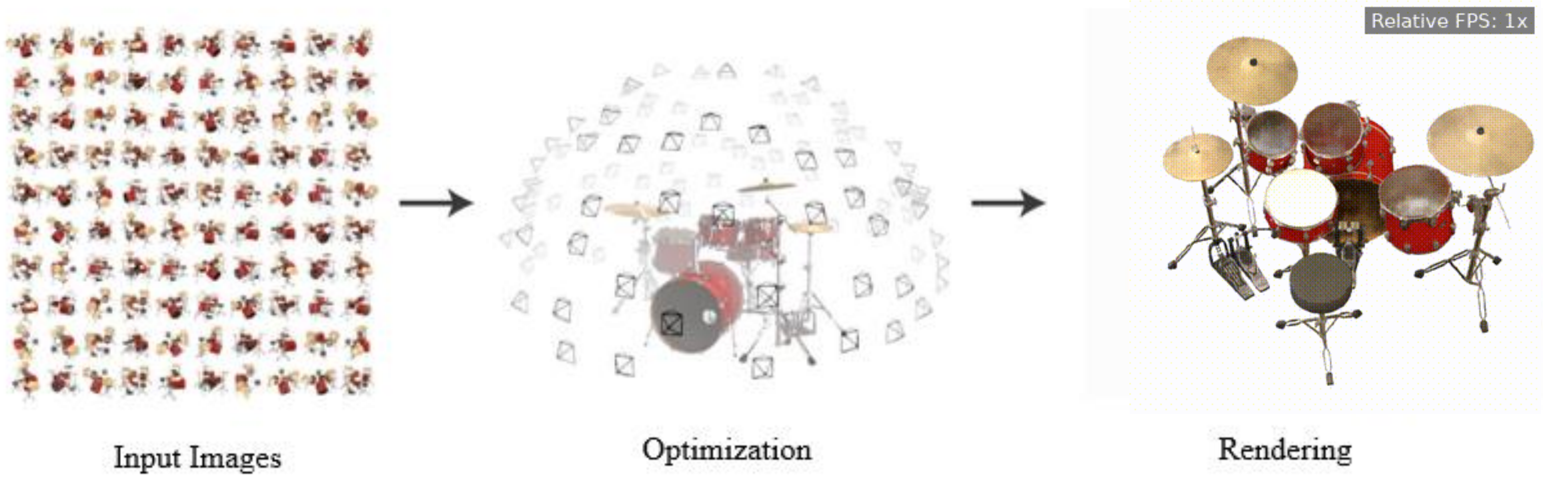
"ghost eating a hamburger"

"a pig wearing a backpack"

We have a small-scale 3D dataset but images on an internet scale.

Images are projections of 3Dfrom specific angles.

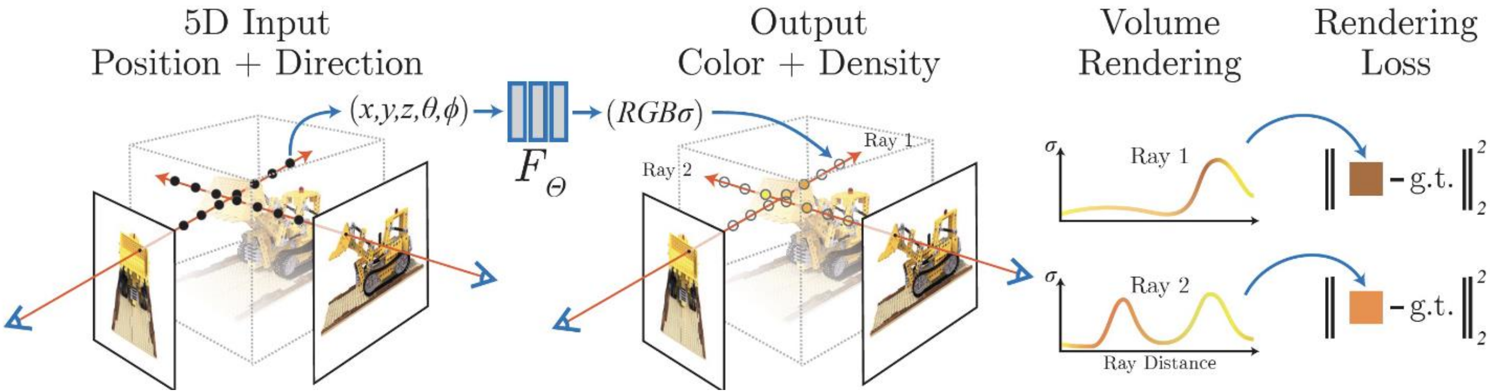
3D Reconstruction by NeRF or Gaussian Splatting



B. Mildenhall, et al., NeRF: Neural Radiance Fields

NeRF/3DGS Optimization

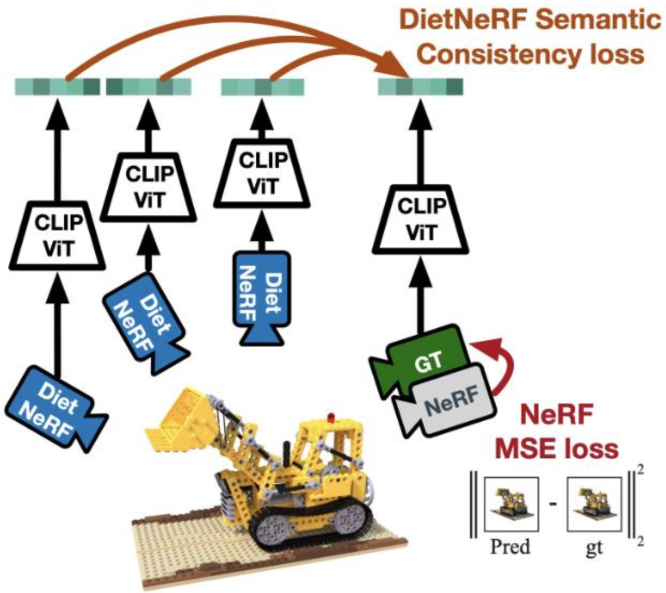
1. *Render a NeRF representation into a specific view.*
2. *Compute the difference with the given image.*
3. *Update the NeRF using gradient descent.*



Can we perform NeRF reconstruction with a few images?

DietNeRF *[Jain et al., ICCV 2021]*

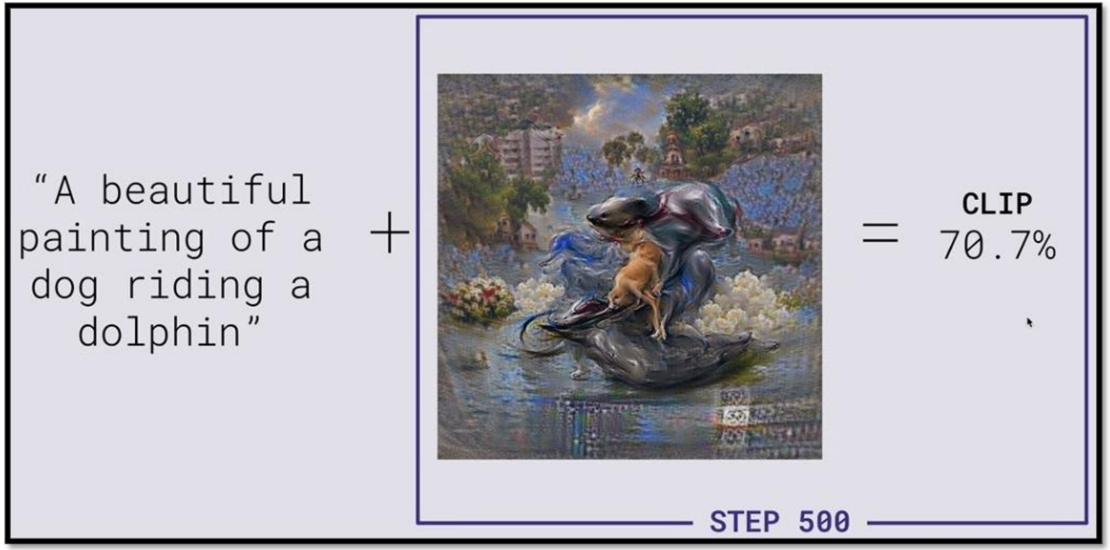
Use priors learned by CLIP *[Radford et al., 2021]*, ***a text-image model.***



"a bulldozer is a bulldozer from any perspective"

CLIP [Radford et al., 2021]

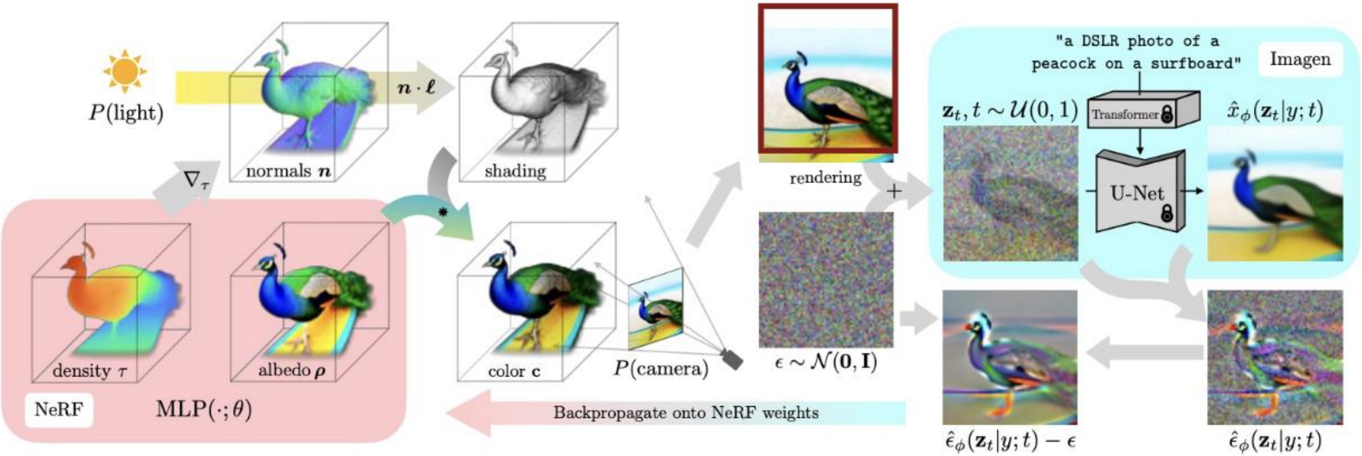
*CLIP takes a text-image pair as input and assesses the **alignment** between the text and the image.*



https://www.youtube.com/watch?app=desktop&v=-b7xKWeADHQ

Knowledge Distillation in 3D Generation

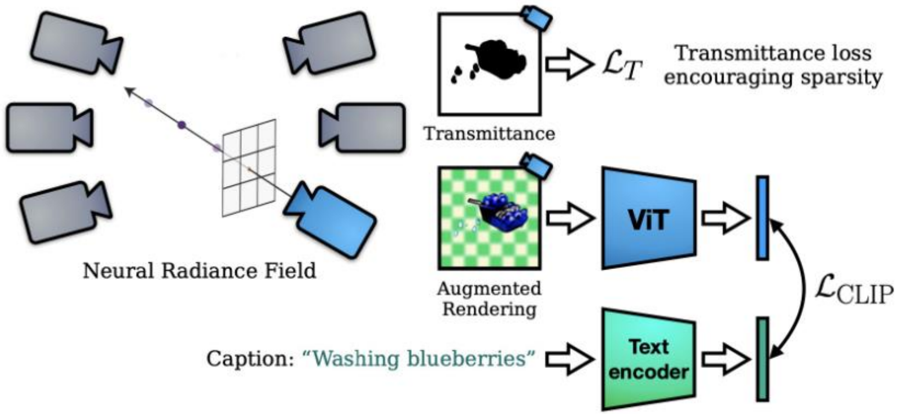
1. *Render the NeRF representation into a specific view*
2. *Compute the alignment to the given text.*
3. *Update the NeRF using the gradient descent.*

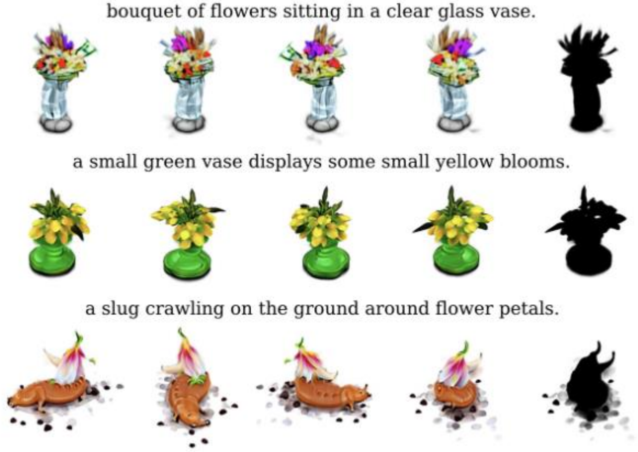


Extreme Case: Zero-Shot NeRF

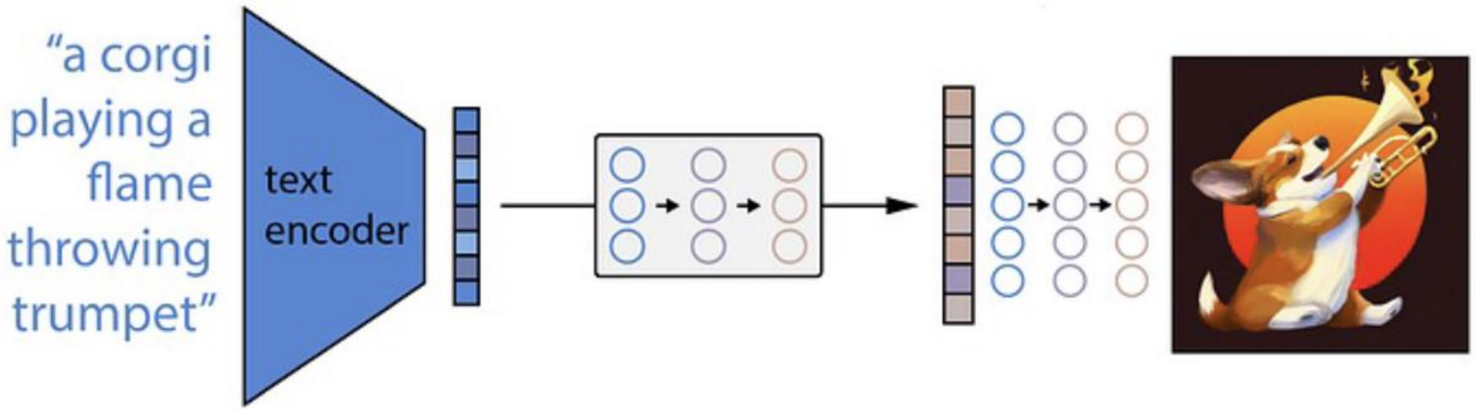
DreamFields *[Jain et al., CVPR 2022]*

Given a text prompt but no images, generate a 3D shape by maximizing similarity between a rendered image and the input prompt in the CLIP embedding space.





Can we use an image diffusion model instead of CLIP?

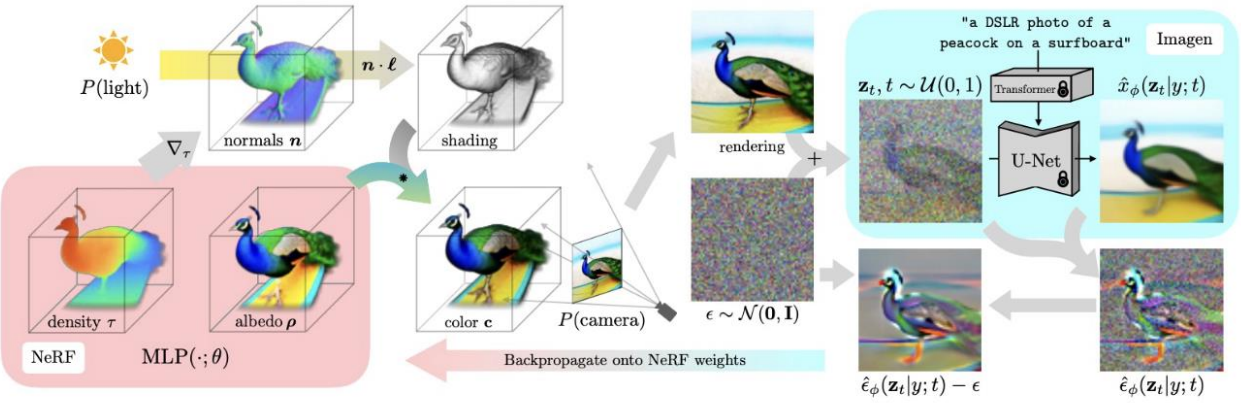


https://medium.com/latinxinai/text-to-image-with-stable-diffusion-4df16da2cfd5

Score Distillation Sampling (SDS)

DreamFusion *[Poole et al., ICLR 2023]*

Proposed the idea of Score Distillation Sampling (SDS), leveraging a pretrained diffusion model to measure the plausibility of rendered images.



Score Distillation Sampling (SDS)

- *How can we utilize a pretrained diffusion model to measure the plausibility of rendered images?*

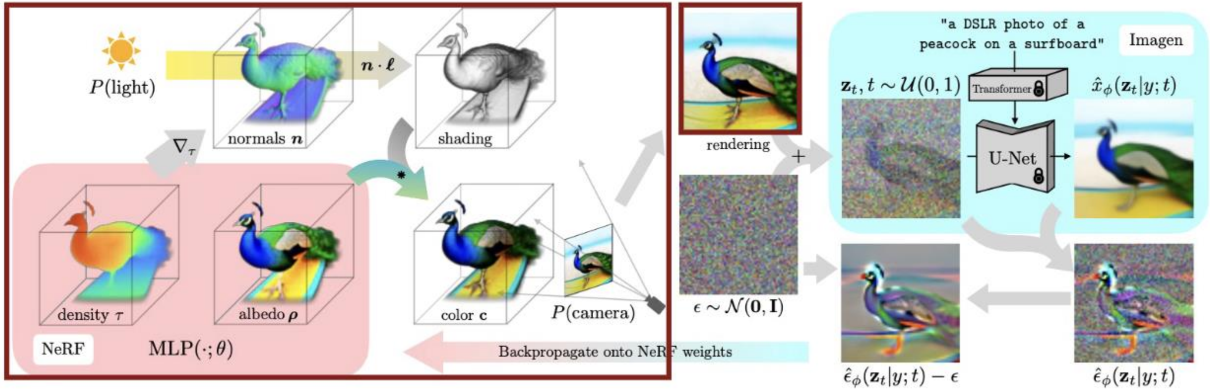
- *Review the loss function:*

$$\mathcal{L} = \mathbb{E}_{t\sim[1,T],\mathbf{x}_0,\boldsymbol{\varepsilon}_t}\left[\left\|\boldsymbol{\varepsilon}_t - \boldsymbol{\varepsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\varepsilon}_t, t\right)\right\|^2\right]$$

If the training of the diffusion model has converged, the loss for real data $\mathbf{x}_0$ will be close to zero.

Score Distillation Sampling (SDS)

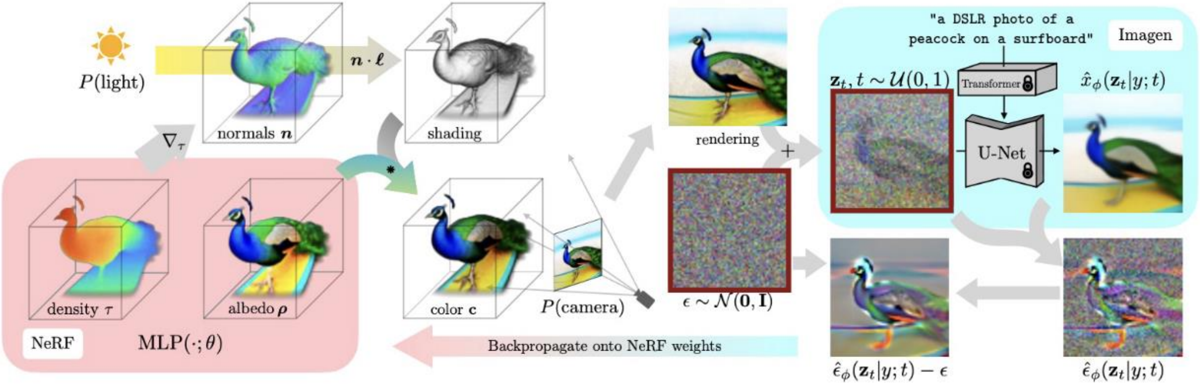
1. ***Render** the NeRF representation into a specific view.*
   Let ϕ denote the NeRF parameter, and
   $\mathbf{x}_0 = g(\phi)$ denote the rendered image.



Score Distillation Sampling (SDS)

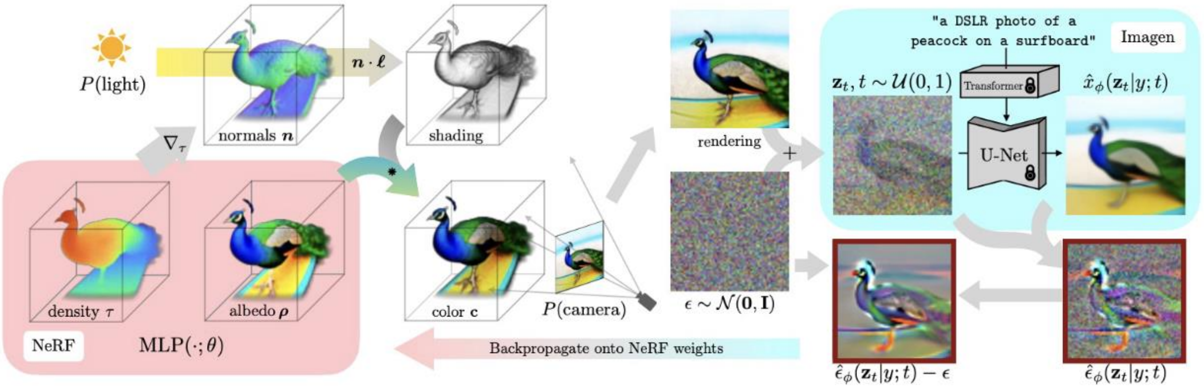
2. *Add noise to the rendered image* $\mathbf{x}_0$*:*

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\varepsilon}_t.$$

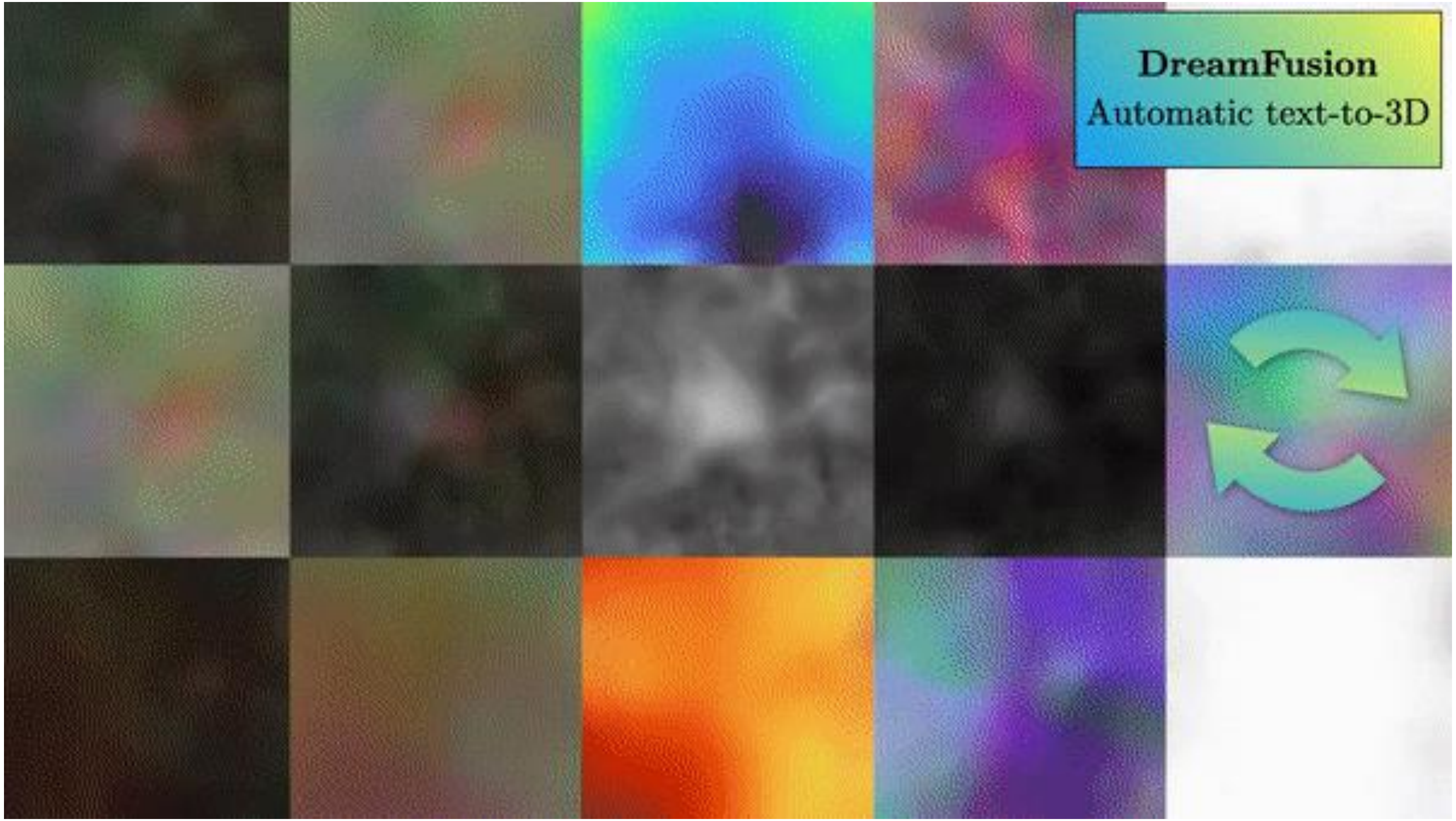


Score Distillation Sampling (SDS)

3. *Perform **gradient descent** on $\mathcal{L}$ with respect to the NeRF parameters ϕ.*

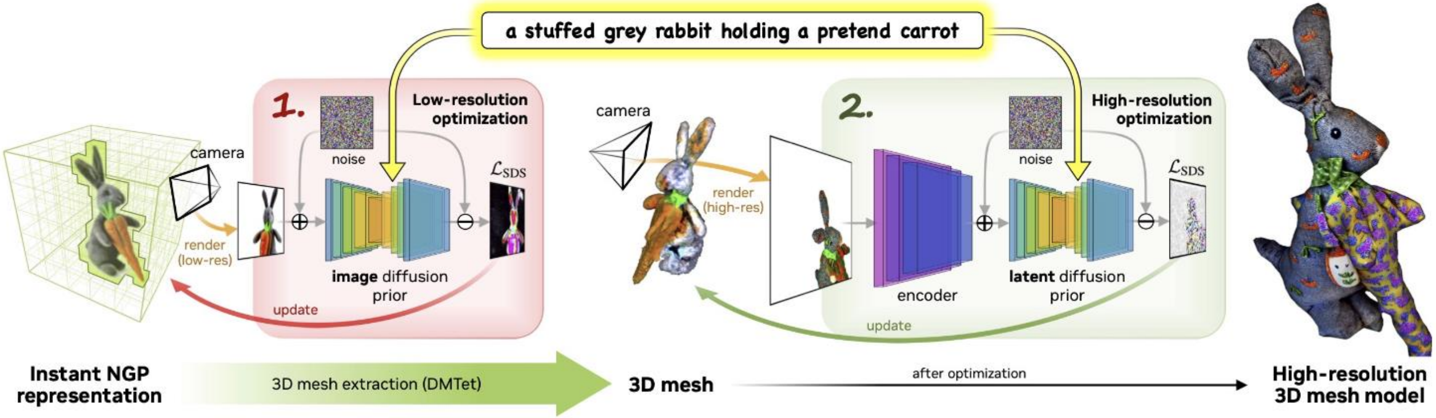


DreamFusion Results



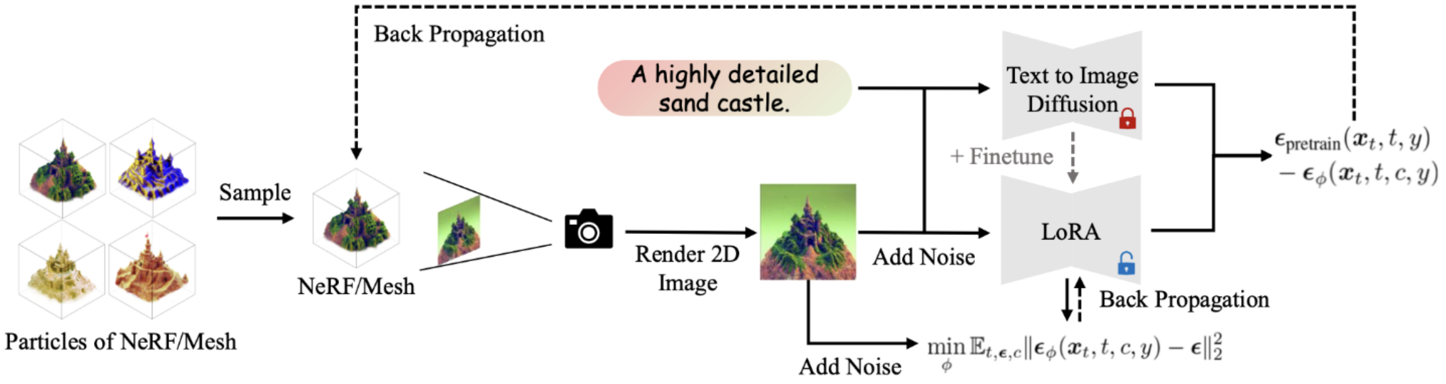
Magic3D [Lin et al., CVPR 2023]

- *Two-stage approach.*
- *Extract a coarse mesh in the first stage, and then texture the mesh in the second stage.*



ProlificDreamer [Wang et al., NeurIPS 2023]

- *Minimize the SDS loss for the **multiple samples** of the NeRF parameters ϕ.*
- *Finetune the diffusion model with the **Low Rank Adaptation (LoRA)** technique.*



Limitation of SDS

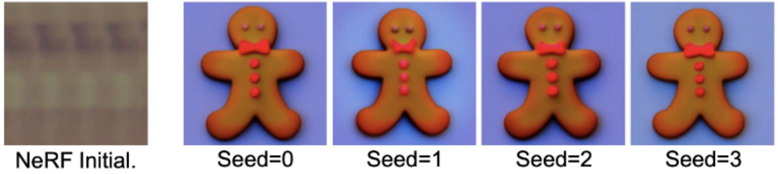
It does not converge well without a high CFG weight (e.g., $w = 400$***) and thus suffers from model collapse.***

"a delicious hamburger"

$w = 400$

$w = 7.5$

Credit: Jaihoon Kim



"gingerbread man"

Huang et al., DreamTime: An Improved Optimization Strategy for Text-to-3D Content Creation, arXiv 2023.

Limitation of 3D Generation from 2D Priors

Twitter @_akhaliq, Prompt: "A DSLR photo of a squirrel"

Limitation of 3D Generation from 2D Priors

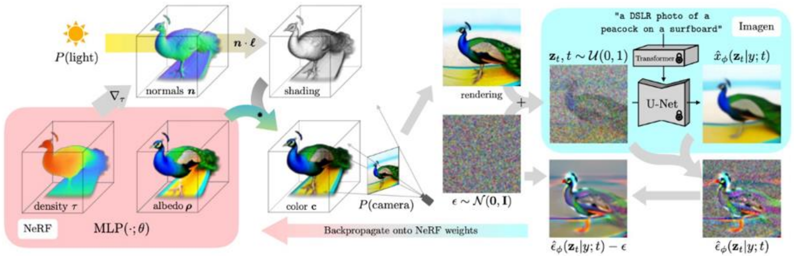
Supervision for geometry is still needed!

ProlificDreamer, Wang et al., 2023.

StableDreamFusion

What’s Next for3D Generative Models?
1. Combining 3D Supervision

While pretrained image generative models will continue to be valuable for 3D generation, the key to producing realistic and solid 3D shapes would lie in utilizing small-scale 3D priors.



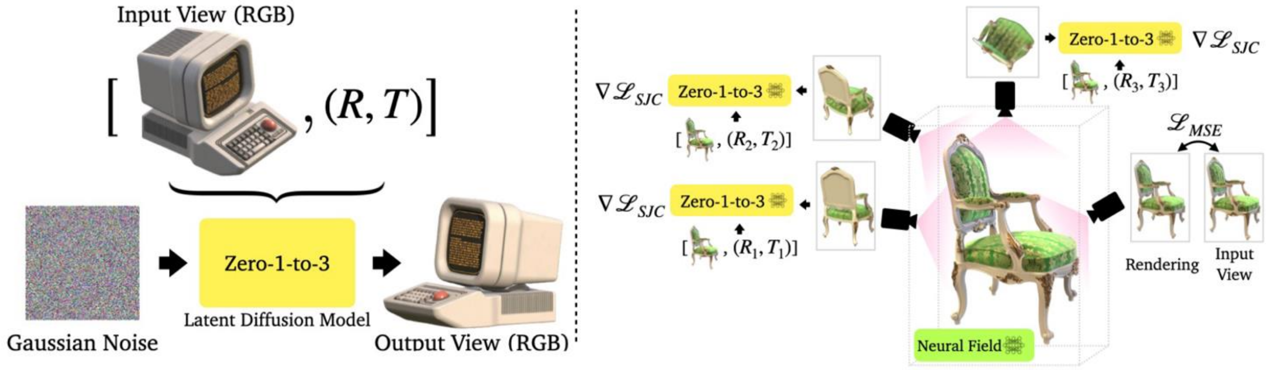
DreamFusion
Google

Objaverse
Allen Institute

Zero-1-to-3 [Liu et al., 2023]

Novel view generation

An image diffusion model generating a novel view image conditioned by another view image and camera pose.

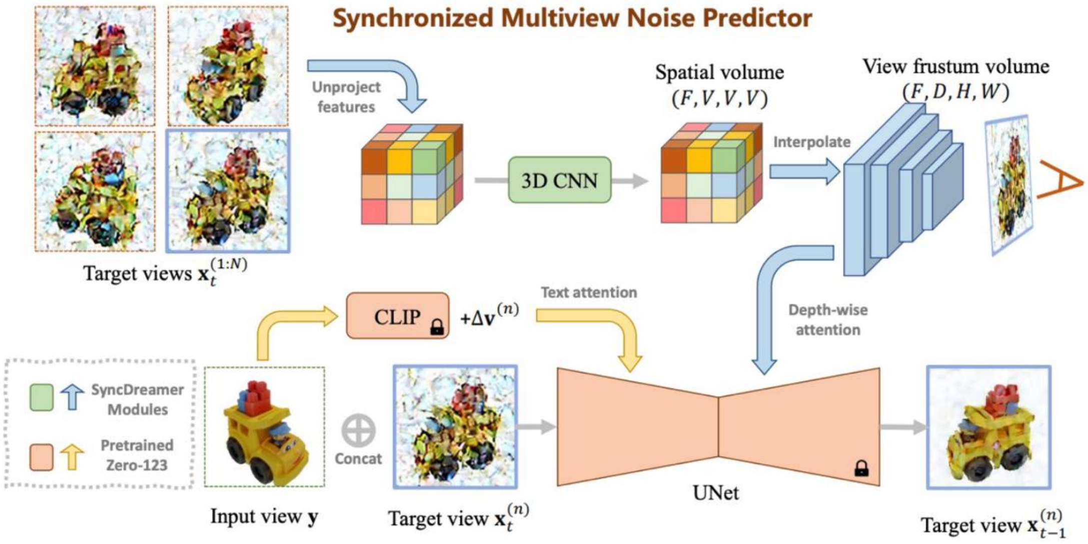


Novel View Synthesis

3D Reconstruction

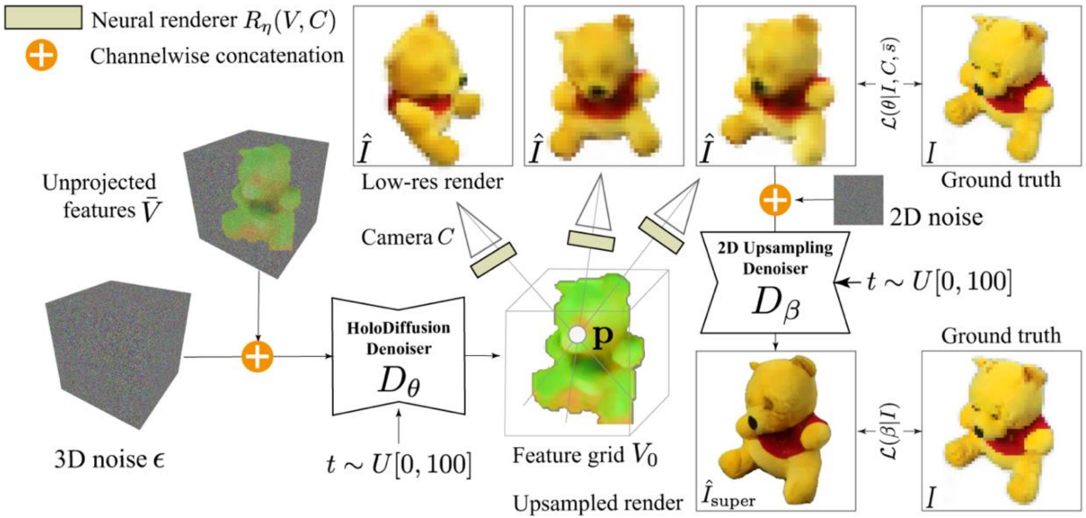
SyncDreamer [Liu et al., 2023]

Utilize Zero 1-to-3 to learn the joint probability distribution of multi-view images.



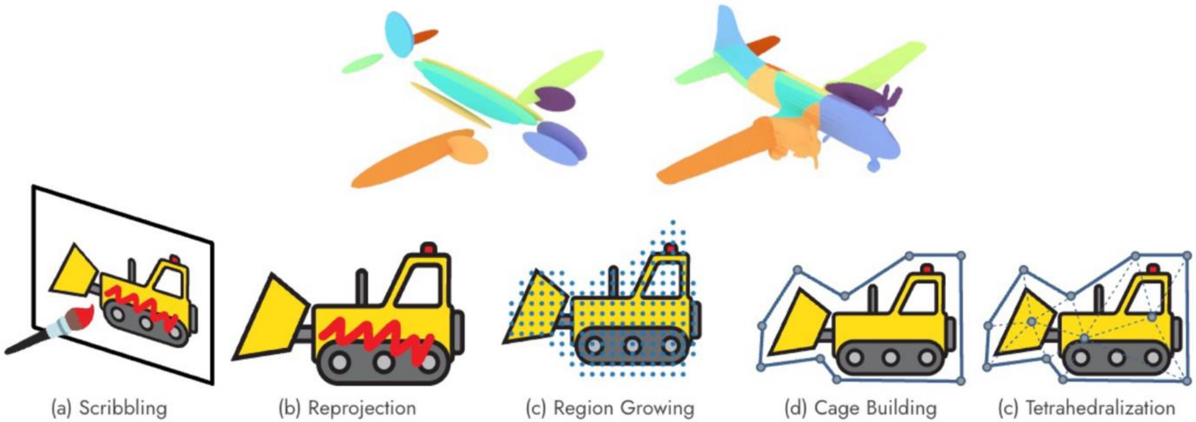
HoloFusion [Karnewar et al., 2023]

- ***Train a 3D diffusion model using multi-view images only.***
- ***Can be extended to integrate 2D priors.***



2. Generation → Editing

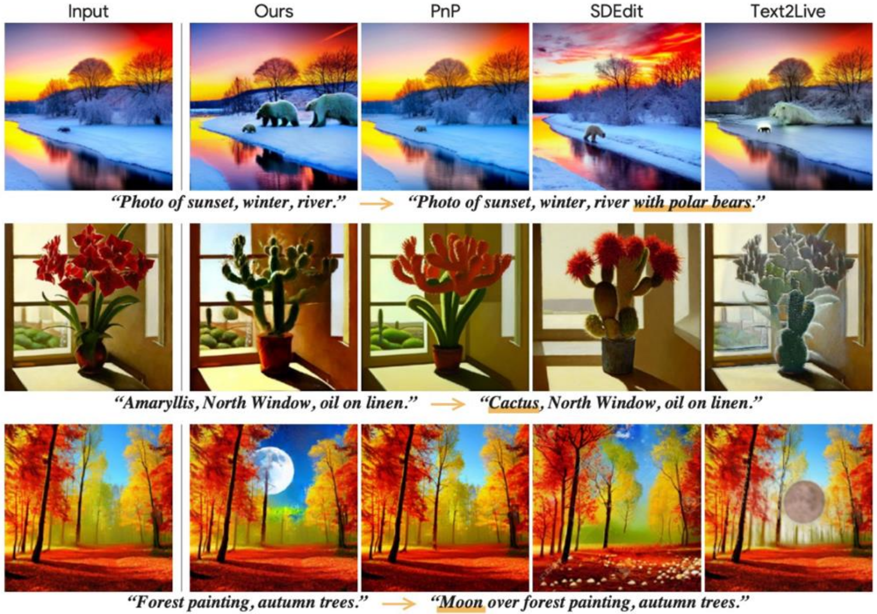
The focus of 3D generative models will shift towards creating versatile models capable of not only generating but editing and manipulating 3D shapes.



NeRFshop, Jambon et al., I3D 2023.

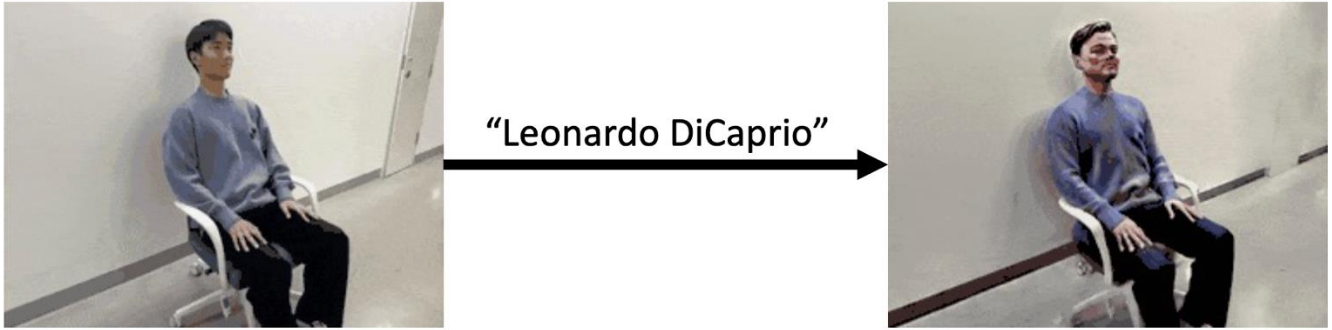
Delta Denoising Score [Hertz et al., 2023]

A new loss function for zero-shot image editing.



Posterior Distillation Sampling [Koo et al., 2024]

A new loss function for zero-shot NeRF editing.



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